

Scenario audit of Cuba’s 2021–2026 population decline

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Scenario ensembles, an infant-mortality consistency check,
and triangulation of independent registers

Supplementary figures (Model E detail; life expectancy; provinces):
`cuba-depopulation-2021-2026-supplement.pdf`

Abstract

Background. Cuba is losing population at one of the fastest peacetime rates in the Western Hemisphere. No census has been completed since 2012, and the national statistics office (ONEI) removed more than one million people from its published headcount in a single 2024 revision. End-2025 published levels still span roughly 8.0–11.0 million. This paper audits the 2021–2025 headcount decline with transparent scenario ensembles, decomposes migration versus internal demographic change, and reports an end-2026 scenario continuation.

Methods. We solve the demographic accounting identity with Monte Carlo simulation ($N = 10^6$) under prior sets labeled Models A–D (scenarios, not identified estimators), with optional Gaussian-copula dependence among baseline overstatement, migration, and death registration. Model D is an *illustrative central scenario*: migration priors comparable to Model B, with residual death under-registration tightened after an infant-mortality consistency check (a Venezuela analogy, not structural identification). External registers are used as ceilings or soft corroboration; Model D is never treated as external validation of itself. Crude registered excess mortality is reported separately from a *provisional* 2019-schedule residual (pending a full ONEI age–sex pipeline).

Results. Across scenarios, cumulative 2021–2025 loss spans roughly **1.8–2.4 million** (the primary uncertainty statement). Under the illustrative central scenario (Model D), living population is near **8.59 million** at end-2025 (median loss **2.21 million, 20.5%** on a corrected 2021 base under fixed priors; within-scenario 90% band for loss roughly 1.84–2.60 million). Under those same Model D priors, emigration accounts for about **91%** of absolute loss. A provisional vital-reconstruction companion (Model E) yields ~ 8.96 million and remains non-preferred. An end-2026 scenario continuation under Model D sits near **8.29 million**.

Conclusions. Depopulation is predominantly migratory in the headcount, reinforced by collapsing births, elevated mortality, and accelerated ageing. Against the *official* 2021 stock, the end-2026 register gap approaches one in four (a different definition from Model D’s -20.5% on the corrected base).

Keywords: Cuba; demography; emigration; natural decrease; infant mortality; fertility; ageing; statistical data quality.

1 Introduction

Counting Cubans has become contested. The last completed census was in 2012; the census planned for 2022 has been repeatedly postponed and is now promised for 2026 [ONEI, 2022]. Without a census, published

population depends on projections and a lagging migration register [Preston et al., 2001, United Nations, Department of Economic and Social Affairs, Population Division, 2024]. In July 2024, ONEI reported an “effective population” of 10,055,968 at end-2023 (about one million below prior published levels), then 9,748,007 (end-2024) and 9,434,593 (end-2025) [ONEI, 2024, 2025].

Three incompatible series coexist (Figure 1a): the United Nations World Population Prospects path near 10.9 million (still assuming net emigration of order 2×10^4 per year) [United Nations, Department of Economic and Social Affairs, Population Division, 2024, Economic Commission for Latin America and the Caribbean (ECLAC), 2022]; ONEI’s revised official series (9.43 million at end-2025); and independent studies placing the living population nearer 8.0–8.6 million [Albizu-Campos Espiñeira, 2023, 2024]. Almost three million people separate the extremes. This paper presents a scenario audit of the 2021–2025 headcount decline with transparent prior ensembles, decomposes migratory and internal components, compares official migration accounting with destination registers, and reports an end-2026 scenario continuation.

Cuba’s fertility transition was already among Latin America’s earliest and deepest [Díaz-Briquets and Pérez, 1982, Hollerbach et al., 1984, Díaz-Briquets, 2014], so below-replacement fertility is not new. What is new is the *speed* of headcount loss under combined emigration and natural decrease, in a regional context where Latin America and the Caribbean still grow overall [Economic Commission for Latin America and the Caribbean (ECLAC), 2022, United Nations, Department of Economic and Social Affairs, Population Division, 2024]. Comparable peacetime contractions elsewhere have been driven primarily by emigration (e.g. Venezuela) [Coleman, 2006, García et al., 2019].

Central thesis. The headcount loss is mostly emigration, but a second “internal” engine (birth decline, excess mortality, and ageing) would sustain contraction even if emigration slowed [Lee and Mason, 2014, Lesthaeghe, 2010]. The two engines are not independent: selective exit of working-age adults and women of reproductive age causes much of the fertility decline, so many “births that do not occur” are a delayed migratory effect [Bongaarts and Sobotka, 2012, Díaz-Briquets, 2014].

2 Methods

The balancing equation is the standard demographic accounting identity [Preston et al., 2001]. Registered ONEI aggregates over 2022–2025 are fixed anchors [ONEI, 2025]: births $B^{\text{reg}} = 325,233$, deaths $D^{\text{reg}} = 502,149$, and identity-consistent net emigration $R = 1,501,706$. Official end-2021 population is $P_{2021}^{\text{off}} = 11,113,215$. Each Monte Carlo draw samples latent corrections and applies

$$P_{2021} = P_{2021}^{\text{off}} - U_0, \quad (1)$$

$$D = D^{\text{reg}} D_{\text{fac}} \text{ (+ optional epidemic pulse in B/C)}, \quad (2)$$

$$M_{\text{net}} = R M, \quad (3)$$

$$\Delta = (D - B) + M_{\text{net}}, \quad (4)$$

$$P_{2025} = P_{2021} - \Delta, \quad (5)$$

where U_0 is end-2021 baseline overstatement (people already effectively abroad but still on the roll), $M \geq 1$ scales ONEI net emigration, and $D_{\text{fac}} \geq 1$ scales registered deaths. Percent loss uses the *corrected* base P_{2021} , not P_{2021}^{off} . We draw $N = 10^6$ scenarios. In Model D, (U_0, M, D_{fac}) may be coupled by a Gaussian copula on latent normals; the numeric ρ matrix is an illustrative dependence assumption (default scientific reading treats medians as driven by margins; an independence sensitivity barely moves them). Within-scenario 90% bands are *prior-predictive* spreads under fixed Beta margins—not sampling error from data. The **range across Models A–D** (~ 1.8 – 2.4 million of cumulative loss) is the primary structural uncertainty

statement [Pérez-Riverol, 2026d,b].

We define five prior sets. **A (conservative)** uses modest U_0 and M near ONEI with a small death multiplier (`model_from_2021.py`). **B (crisis-adjusted)** raises M , allows death under-registration up to $\sim 14\%$, and adds an explicit epidemic pulse. **C (upper stress)** pushes U_0/M and under-registration further (cap $\sim 18\%$) with a large elderly mortality pulse. **D (illustrative central)** keeps migration priors comparable to B but *tightens* death under-registration after the infant-mortality consistency check (Table 1). **E (provisional companion)** is a vital reconstruction via a health-system deterioration score (supplement) and is not preferred for public claims. Models A–D are **scenarios** (prior narratives), not scientifically identified estimators of a unique living population. Model D is preferred as an illustrative central path inside the across-scenario envelope because its migration prior matches B while replacing B’s wide mortality pulse with a sentinel-bounded D_{fac} —not because M is uniquely identified by a likelihood. Models B and D yield nearly identical headcount medians because migration priors dominate absolute loss. ONEI’s 2023–2024 revisions already absorbed a large lagged stock; scaling revised R by $M > 1$ and subtracting large U_0 can double-count the same outdated-register defect. We therefore always report the ONEI-anchored neighbourhood (Model A / $M \approx 1$) beside Model D, and treat $U_0 + (M - 1)R$ as constrained by destination floors plus documented lag—not as unbounded free parameters.

Table 1: Model D prior specification (margins; optional Gaussian copula). Beta scales: $U_0 \in [0, 700,000]$, $M \in [1, 1.72]$, $D_{\text{fac}} \in [1, 1.09]$. Birth noise independent $B = B^{\text{reg}}\mathcal{N}(1, 0.01^2)$.

Symbol	Margin	Interpretation
U_0	$700,000 \times \text{Beta}(2.0, 2.4)$	End-2021 roll overstatement
M	$1 + 0.72 \times \text{Beta}(2.2, 2.4)$	Multiplier on ONEI net emigration R
D_{fac}	$1 + 0.09 \times \text{Beta}(1.8, 3.2)$	Residual death under-registration (at-home elderly channel)

Infant mortality rose from 5.0 to 9.9 per thousand live births (2019–2025), a $\sim 98\%$ increase. Infant deaths are comparatively hard to hide (hospital occurrence; international scrutiny), so IMR is treated as a high-integrity sentinel of health-system *stress*, not as a calibrator of elderly death completeness. As a loose foreign analogy (not a Cuban structural parameter), Venezuela’s crisis saw IMR rise $\sim 76\%$ (roughly 2012–2017) alongside an all-cause crude death-rate rise of order $\sim 25\%$ [García et al., 2019, García and Aburto, 2019, Calibration note, 2017]. A log-elasticity reading is

$$e = \frac{\ln(1.25)}{\ln(1.76)} \approx 0.39, \quad (6)$$

so Cuba’s IMR rise would predict $(1.98)^{0.39} - 1 \approx 31\%$ higher CDR if the analogy transferred. Registered Cuban CDR rose from 9.7 to 12.9 per thousand by 2024 ($\sim 33\%$), roughly consistent with that order of magnitude [ONEI, 2025]. This remains a **consistency check against very large hidden institutional undercounts**, not identification of D_{fac} ; residual under-registration in Model D is still limited to a small at-home elderly channel (central $\sim 3\%$, cap 9% ; Table 1) [Pérez-Riverol, 2026b].

Two distinct mortality quantities are reported and must not be added or substituted [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018]. For crude registered excess, on crisis-adjusted mid-year stocks $\tilde{P}_t$, we define a no-crisis counterfactual CDR with a mild ageing drift from the 2019 level 9.7 per thousand, $\text{CDR}_t^{\text{cf}} = 9.7 + 0.16(t - 2019)$, and

$$E_t^{\text{crude}} = D_t^{\text{reg}} - \text{CDR}_t^{\text{cf}} \cdot \tilde{P}_t / 1000. \quad (7)$$

Summing 2020–2025 yields $\sum E_t^{\text{crude}} \approx 153,000$ (of which $\sim 66,000$ in 2024–2025); most of this toll is already inside registered death totals, and the linear CDR drift is a transparent baseline, not full age standardization. Separately, Figure 4a compares registered deaths to *illustrative* expected deaths under a provisional 2019 schedule bridge ($\sim 95\text{k}$ and $\sim 92\text{k}$ versus registered 128.1k and 136.2k), implying a combined residual of order $\sim 77,000$ for 2024–2025 (`scripts/mortality-schedule-residual.py`). A wider 65,000–90,000 band is retained as sensitivity to population-stock choice pending a full ONEI table 3.15 age–sex application to explicit $P_{a,t}$ [Preston et al., 2001]. We deliberately avoid calling this a Kitagawa decomposition: Kitagawa separates rate versus composition contributions to a rate difference; here we report a direct expected-deaths residual [Kitagawa, 1955]. Mortality rates at ages 60+ remaining 11–27% above 2019 are consistent with elevated age-specific risks, but that claim also awaits full schedule publication.

Published population levels are grouped by information content (Figure 2). Ceilings that do not purge emigrants include the UN World Population Prospects, the MINSAP denominator, and the electoral roll (soft upper bounds only). The official ONEI revised headcount is 9.43 M at end-2025. Exit-correcting or independent constructs include housing $\times$ occupancy (~ 8.7 M; fragile occupancy scalar) and Albizu-Campos 2023/2024 (8.0–8.6 M; methods that share migration-critique assumptions with this paper). Model D (8.59 M) is the study estimand and is *not* external validation of itself; agreement with Albizu/housing is partly shared-prior echo, while ceilings bound the upside. The soft band from exit-correcting sources is near 8.0–8.9 million.

The settled-concept destination floor (Table 2) comprises United States $\sim 800,000$ (administrative/CBP-pipeline statuses, *not* ACS stock), non-U.S. total $\sim 250,000$ (Spain $\sim 135,000$ + Uruguay $\sim 35,000$ + rest $\sim 80,000$), and floor ~ 1.05 million [Pérez-Riverol, 2026a, Moslimani and Passel, 2024]. ACS Cuban-born stock rose by only $\sim 335,000$ from 2019–2024—a lower stock-change floor that must not be silently equated with the 800,000 administrative figure. Encounter counts and nationality applications are separate concepts and are never added into the settled floor [U.S. Customs and Border Protection, 2024].

Table 2: Bridge from settled destination floor toward scenario migration engines. Only the floor is ledger-like; later rows are residuals or priors, not destination microdata.

Concept	Approx. count	Role
Settled destinations floor (US admin + non-US)	1.05 M	Lower bound (settled concept)
U.S. ACS Cuban-born rise 2019–24 (not settled)	~ 0.34 M	Stock-change floor; do not add
Unquantified gap toward ONEI R	$\rightarrow \sim 1.5$ M	Residual, not validated persons
ONEI identity-consistent net R (2022–25)	1.50 M	Official flow after revision
Albizu-Campos-type net	~ 1.8 M	Independent upper narrative
Model D median $R \times M$	~ 2.0 M	Prior-driven scenario engine
U.S. encounters 2022–2024 (not settled)	~ 0.85 M	Events; do not add to floor
Spain nationality applications	~ 0.30 M	Many applicants still in Cuba

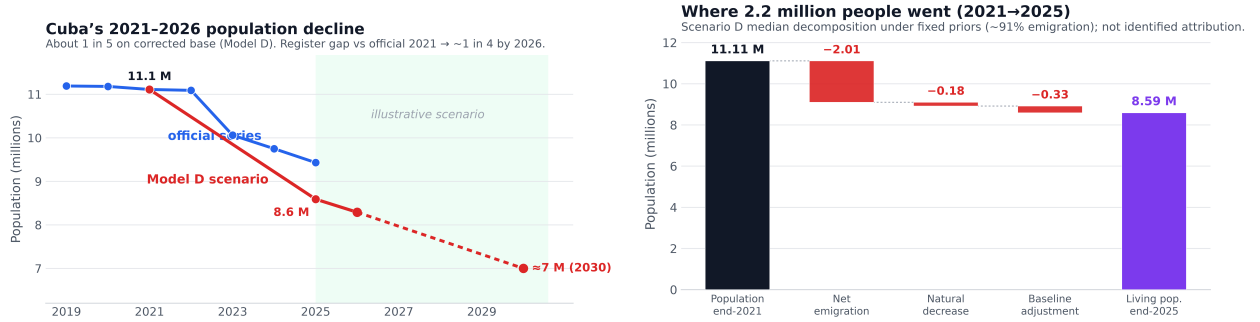
Destination data do *not* identify Model D’s ~ 2.0 million net emigration; they support a large floor and leave the 1.05 $\rightarrow$ 2.0 path as soft gap-filling plus a migration prior. If true net emigration were nearer 1.5 than 2.0 million, living population would sit toward ~ 8.9 rather than ~ 8.6 million; cumulative loss would remain large and emigration would still dominate absolute loss under every scenario in Table 3. Conditional on end-2025 draws, the end-2026 scenario continuation uses births centered at 65,682 ($\mathcal{N}$ noise 3%), deaths centered at $1.04 \times 136,214$ times the same D_{fac} , and a three-regime net emigration mixture with probabilities (0.35, 0.35, 0.30) on centers (150,000, 210,000, 300,000) plus 15% log-scale noise. The $\sim 4\%$ death lift and regime weights are scenario assumptions, not a validated forecast. Paths to ~ 7 million by 2030 in Figure 1a are **illustrative trend scenarios** (visually the post-2026 segment); they are not probabilistic forecasts.

3 Results

Table 3 summarizes end-2025 medians for Models A–E. The scientifically primary uncertainty statement is the across-scenario loss envelope (~ 1.8 – 2.4 million). Illustrative Model D yields median population **8.59 million** and loss **2.21 million** (20.5% on the corrected base under fixed priors; within-scenario 90% band 1.84–2.60 million). Under Model D priors, migration’s share of absolute loss is about **91%**. Model A (ONEI-anchored, $M \approx 1$) remains the conservative floor neighbourhood (~ 9.16 M). Provisional Model E yields ~ 8.96 M (loss 2.13 M): same loss band, higher stock, partly because its U_0 prior is much smaller than Model D’s; an adversarial gate does not promote E over D [Pérez-Riverol, 2026c]. Percentage language: against official 2021 (11.11 M), end-2025 register-gap loss is roughly 20–23%; against the corrected study base it is about 16–23% across A–D. A 2026 scenario continuation places Model D near **8.29 million** (further loss $\sim 300,000$; across-scenario band roughly 7.9–8.9 million).

Table 3: Models A–E: population loss 2021–2025 and end-2025 population (medians). A–D are scenarios; D is the illustrative central path; E is provisional.

Model	Main assumption	Loss (M)	%	Pop. end-2025 (M)
A · conservative	ONEI-anchored	1.80	16.4	9.16
B · crisis-adjusted	under-reg. + independent migration	2.19	20.4	8.57
C · upper stress	analog upper bound	2.41	22.6	8.26
D · sentinel (illustrative)	IMR check + migration priors near B	2.21	20.5	8.59
E · vital reconstr. (prov.)	HSDS $\rightarrow B^*, D^*$ (differs in U_0)	2.13	19.2	8.96



(a) Official versus scenario living population, 2021–2030 (b) Accounting decomposition under Model D medians (Model D illustrative path in red; post-2026 segment is a (2021–2025); illustrative scenario, not identified attribution-non-probabilistic trend scenario).

Figure 1: Headline population path and accounting decomposition.

Sources that do not purge emigrants (electoral roll, MINSAP health denominator) sit high and function as soft ceilings. Exit-correcting constructs (housing occupancy ~ 8.7 M; Albizu-Campos 8.0–8.6 M) sit near Model D but share migration-critique assumptions with this study; they are not independent censuses. The official 9.43 M sits between the two families (Figure 2). Voter turnout fell sharply between 2018 and 2023; that pattern is compatible with physical absence but also with abstention, logistics, and political factors, and is *not* used as a person count.

Table 2 separates settled counts from encounters, ACS stock change, and nationality applications. The settled floor (~ 1.05 M) confirms that emigration is large while remaining a lower bound. Closing the gap to ONEI’s ~ 1.5 million is an unquantified residual; Model D’s ~ 2.0 million is a prior on R , not destination identification. If true net is nearer 1.5 than 2.0 million, living population sits toward the high end of our band (~ 8.9 rather than 8.6 M). Emigration still dominates absolute loss under every scenario in Table 3.

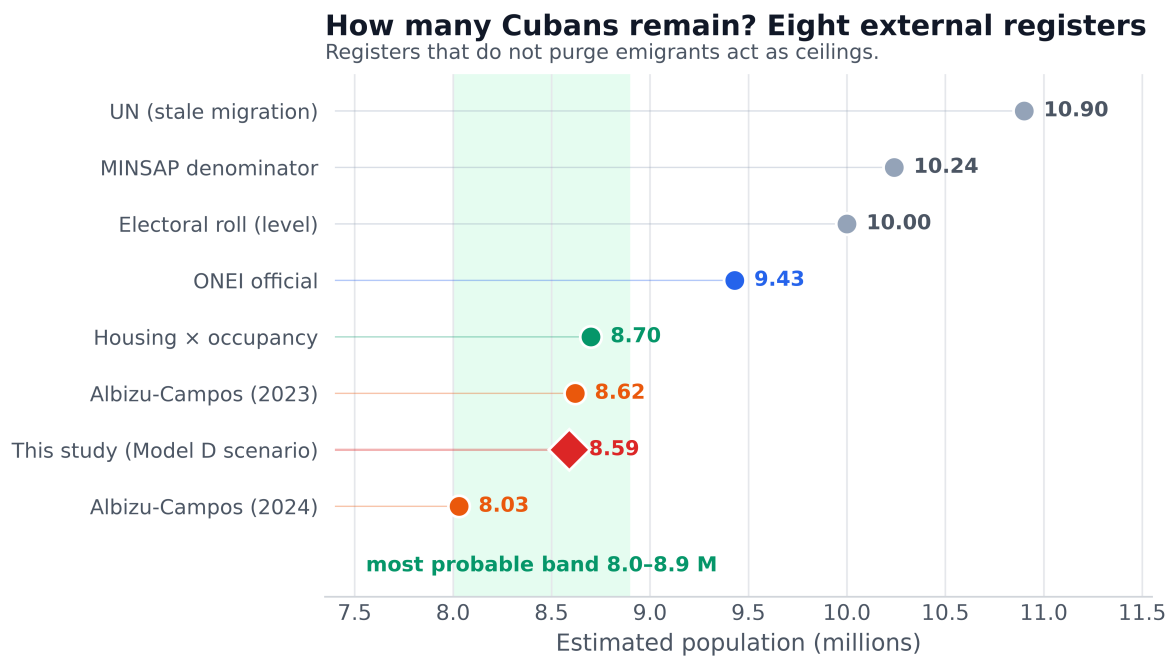
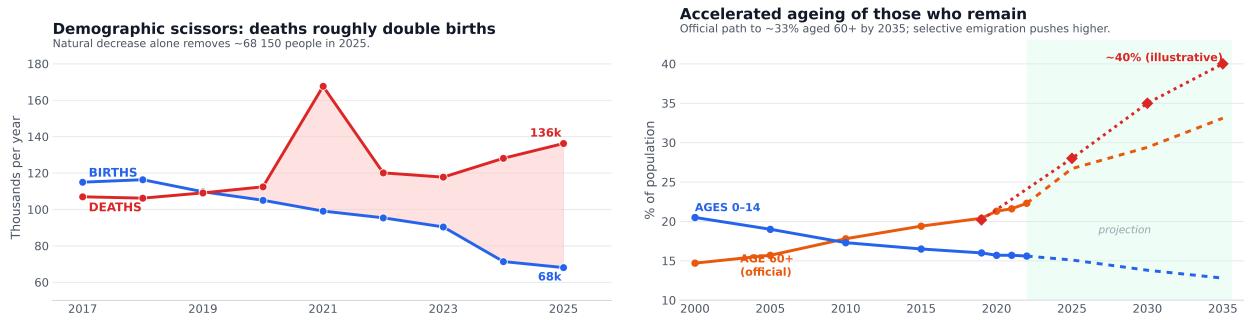


Figure 2: Triangulation of end-2025 population levels. Model D is the study estimand (not a validator); Albizu/housing are soft, assumption-laden anchors.

Reducing the crisis to emigration alone is incomplete: emigration can reverse, but missing births and an aged residual population cannot. A useful distinction is **realized loss** (exit-dominated) versus **future inertia** (natural decrease increasingly important). In 2019 the natural balance was still slightly positive (+636); in 2025 deaths exceeded births by **68,150** (68,064 births versus 136,214 deaths; Figure 3a). That 2025 deficit matches the 2021 COVID peak without a pandemic: natural decrease is now structural. The total fertility rate fell to 1.29 children per woman in 2025, the lowest since 1958 and well below replacement (2.1) [Díaz-Briquets, 2014, Bongaarts and Sobotka, 2012, Lesthaeghe, 2010]. Emigration removes young women (majority-female flow) and those who remain defer births under scarcity. Births fell 38% (from ~110,000 in 2019 to 68,000 in 2025). A back-of-envelope split into “half fewer women / half lower TFR” is *illustrative only*: ONEI TFR denominators may themselves be inflated by absent women, which would overstate the rate half; a formal exposure–ASFR decomposition is not claimed here. Official projections move the share aged 60+ from 20.4% (2019) to 26.7% (2025) and toward 33% by 2035 under a larger living stock than Model D [ONEI, 2025, Lee and Mason, 2014]. Under a smaller, youth-depleted stock the elderly share would be higher; any ~29% / ~40% figures in companion graphics are **illustrative sketches**, not cohort-component estimates. Median age reached **45** years in 2025 on official reporting (Figure 3b).

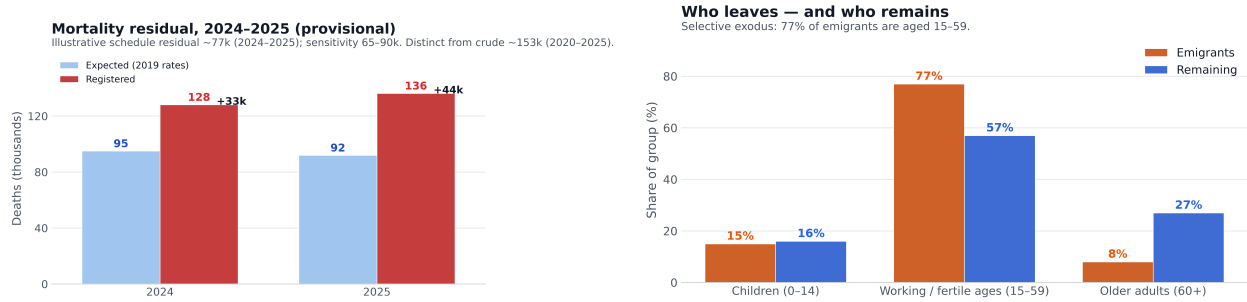


(a) Births versus deaths (“scissors”): deaths roughly double births by 2025. **(b)** Accelerated ageing as working-age adults leave (official structure).

Figure 3: Internal demographic engine: natural decrease and accelerated ageing.

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20; an independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023, Brønnum-Hansen and Albizu-Campos Espiñeira, 2023]. That decline is large relative to COVID-era international losses [Aburto et al., 2022, Msemburi et al., 2023]; LE re-expresses mortality rather than independently corroborating Model D (series in the supplement). Infant deaths rose from 461 (2018) to 715 (2022); the rate reached 9.9 per thousand in 2025 (+98% since 2019). Figure 4a reports a *provisional* 2019-schedule residual of order $\sim 77,000$ for 2024–2025 combined (sensitivity band 65,000–90,000; coarse bridge in `scripts/mortality_schedule_residual` distinct from crude registered excess of $\sim 153,000$ over 2020–2025 [Karlinsky and Kobak, 2021]).

About **77%** of emigrants are aged 15–59; nearly half are under 35 (Figure 4b). Among those remaining, official structure places about 26.7% at ages 60+ (median age ~ 45)—not a Model D-consistent stock. Exit hollows the centre of the pyramid and removes those who would have children [Coleman, 2006, Lee and Mason, 2014]. The flow is majority female [Albizu-Campos Espiñeira, 2023] and concentrated at working ages (Figure 4b). Subnational outflow is uneven: ONEI-reported Havana net emigration is about **–33 per 1,000** in 2025 (provincial ranks more reliable than levels if national migration accounting is biased; map in the supplement). Together, mortality and selectivity close the internal story of who dies, who leaves, and who remains (Figure 4).



(a) Registered deaths versus a provisional 2019-schedule expectation (illustrative). Combined residual $\sim 77k$ for 2024–2025; sensitivity band 65–90k pending full age–sex computation. Crude registered excess $\sim 153,000$ over 2020–2025 is a separate quantity.

Figure 4: Mortality residual (provisional) and selectivity of exit. Life expectancy and provincial maps appear in the supplement.

Regionally, Latin America and the Caribbean still grow ($\sim +0.7\%$ per year; Table 5, UN WPP / CEPAL approximations). Cuba shows net population loss of about -3.3% annually on recent official-path comparisons. Among peacetime emigration-driven contractions, Venezuela is the closest regional peer (Table 4); disaster-driven cases are not treated as peers. By **end-2026**, the Model D scenario continuation implies about 300,000 further loss: living population near **8.29 million** (illustrative across-scenario band 7.9–8.9 M). Against the official 2021 base, the register gap approaches ~ 2.8 million ($\sim 25\%$): a different estimand from Model D’s -20.5% on the corrected base. Illustrative trend paths toward ~ 7 million by 2030 (Figure 1a, post-2026 segment) are scenarios only.

4 Discussion

Model D is retained as the illustrative central scenario because the IMR consistency check shrinks the need for large hidden-death multipliers and focuses residual uncertainty on migration [García et al., 2019, Pérez-Riverol, 2026b]. It is not an identified point estimate: absolute loss remains $\sim 91\%$ migratory under D’s priors, and U_0 with $M > 1$ can share a common outdated-register defect. Provisional Model E lands in the

Table 4: Cuba versus selected peacetime emigration-driven contractions. Cuba row uses Model D on corrected base; comparators are approximate and not window-harmonised. Puerto Rico row confounded by hurricane disaster.

Case	Cum. loss	Approx. annual	Dominant engine
Venezuela 2013–24	$\sim 25\%$	1–3%	emigration
Puerto Rico post-2017 (María era)	large net exit	—	emigration + disaster
Cuba 2021–25 (Model D)	$\sim 20.5\%$ (corrected base)	$\sim 3\text{--}5\%$	emigration + internal

Table 5: Annual population growth: Cuba versus the region (approximate; UN WPP 2024 revision and CEPAL, rounded).

Country / region	Annual growth
Guatemala	+1.7%
Bolivia	+1.4%
Dominican Republic	+1.0%
Mexico	+0.8%
Colombia	+0.7%
<i>Latin America & Caribbean</i>	+0.7%
Brazil	+0.4%
Chile	+0.3%
Uruguay	+0.1%
Venezuela	$\approx 0\%$ (after $\sim 25\%$ loss 2013–20)
Cuba	–3.3%

same loss band with a higher stock (~ 8.96 M) and is summarized in Table 3 and the supplement [Pérez-Riverol, 2026c]; its adversarial gate (different U_0 , ageing double-count risk in S_t , migration still identifying P^*) keeps E non-preferred. Within-scenario intervals look falsely precise relative to the across-scenario envelope—the envelope should lead. The scientific lead is Model D $\sim 20.5\%$ loss by end-2025 on the *corrected* base (about one in five); as a register-gap aside, against official 2021 (11.11 M), cumulative gap by end-2026 approaches $\sim 25\%$ (about one in four). These are not interchangeable headlines. Exit is age- and sex-selective; it ages the island and structurally depresses fertility [Coleman, 2006, Díaz-Briquets, 2014, Lee and Mason, 2014]. The natural balance, positive in 2019, fell to $-68,150$ in 2025; the provisional schedule residual in 2024–2025 is consistent with elevated mortality beyond ageing alone, pending full age–sex computation [Karlinsky and Kobak, 2021, Msemburi et al., 2023]. WFP food-insecurity and anaemia indicators, and the fall in agricultural GDP, are discussed only as *contextual* amplifying pathways [World Food Programme, 2024]; they do not enter D_{fac} or excess formulas and are not identified causes of the death or birth series. ONEI’s declared 2022 net migration (+991) is inconsistent with hundreds of thousands of U.S. Cuban arrivals that year at any plausible encounter-to-emigrant conversion (encounters are not unique net emigrants 1:1). The accounting relocates much of the exodus into a single 2023 revision jump. Even after that revision, 2024–2025 net balances remain only slightly above U.S.-only recorded entries, while the 2025 balance admits 317,320 gross external emigrations (net $-245,264$): continued understatement of exit. This paper therefore treats the official headcount as a ceiling and contrasts it with exit-correcting sources (Figure 2).

5 Limitations

No post-2012 census; ONEI’s ~24-month emigration rule lags recent exits; destination microdata are incomplete; U.S. “settled” administrative totals are not ACS stock; the IMR analogy is not a Cuban structural parameter; housing-occupancy assumptions are fragile; ageing sketches are not cohort-component projections; life expectancy re-expresses mortality; 2026–2030 paths are scenarios. U_0 and M can double-count register lag; destination encounter ceilings can inflate unique persons; within-scenario intervals give false precision relative to structural uncertainty. The provisional 2019-schedule residual is not yet a published age–sex computation from ONEI 3.15. Model E’s higher population partly reflects a different U_0 prior. Food-insecurity indicators are contextual only.

6 Conclusions

Across scenarios, Cuba’s 2021–2025 population loss lies near 1.8–2.4 million (primary statement). Under illustrative Model D priors, living population is near 8.59 million at end-2025 (not the official 9.43 million), with median loss 2.21 million (20.5% on the corrected base) and emigration about 91% of absolute loss; Model A remains the ONEI-anchored floor neighbourhood. By end-2026 the Model D scenario continuation is near 8.29 million; the register gap against the official 2021 base approaches one in four—a different definition from –20.5%. A provisional schedule residual of order ~77,000 deaths in 2024–2025 (sensitivity 65–90k) is secondary to the migration scenario envelope and must be kept distinct from crude registered excess of ~153,000 over 2020–2025. A 2026 census conducted under standard practice would be informative; the working scenario band is 8.6–9.0 million rather than the 10.9 million World Population Prospects path under still-low net-emigration assumptions [United Nations, Department of Economic and Social Affairs, Population Division, 2024].

7 Data and code availability

ONEI tables and a machine-readable claim sheet are in the public repository <https://github.com/ypriverol/cubascience> under `demographics/`. Interactive companion: <https://ypriverol.github.io/cubascience/demographics/>. Monte Carlo scripts (`scripts/model_*.py`) write summaries to `artifacts/`. Supplementary PDF: `manuscript/en/cuba-depopulation-2021-2026-suppl`. Riverol, 2026c]. Third-party literature is cited, not redistributed.

Competing interests

None declared.

References

- José Manuel Aburto, Jonas Schöley, Ilya Kashnitsky, Luyi Zhang, Charles Rahal, Trifon I. Missov, Melinda C. Mills, Jennifer B. Dowd, and Ridhi Kashyap. Quantifying impacts of the COVID-19 pandemic through life-expectancy losses: A population-level study of 29 countries. *International Journal of Epidemiology*, 51(1):63–74, 2022. doi: 10.1093/ije/dyab207.
- Juan Carlos Albizu-Campos Espiñeira. Independent estimates of cuba’s resident population using electoral rolls. Technical report, Working paper / report, 2023.

- Juan Carlos Albizu-Campos Espiñeira. Cuba: emigración, vaciamiento demográfico y población, 2024. Technical report, Working paper / report, 2024.
- John Bongaarts and Tomáš Sobotka. A demographic explanation for the recent rise in European fertility. *Population and Development Review*, 38(1):83–120, 2012. doi: 10.1111/j.1728-4457.2012.00473.x.
- Henrik Brønnum-Hansen and Juan Carlos Albizu-Campos Espiñeira. Life expectancy in cuba. *Journal of Population Research*, 2023. doi: 10.1007/s12546-023-09296-w.
- Calibration note. Venezuela collapse analog for IMR–CDR elasticity, 2017. Loose analogy only. If IMR rose 76% with CDR +25%, log elasticity $e = \ln 1.25 / \ln 1.76 \approx 0.39$. Applied to Cuba IMR +98%: $(1.98)^{0.39} - 1 \approx 31\%$. Not a Cuban structural parameter; consistency check only.
- David Coleman. Immigration and ethnic change in low-fertility countries: A third demographic transition. *Population and Development Review*, 32(3):401–446, 2006. doi: 10.1111/j.1728-4457.2006.00125.x.
- Sergio Díaz-Briquets. Accounting for recent fertility swings in cuba. *Population and Development Review*, 40(4):727–742, 2014. doi: 10.1111/j.1728-4457.2014.00006.x.
- Sergio Díaz-Briquets and Lisandro Pérez. Fertility decline in cuba: A socioeconomic interpretation. *Population and Development Review*, 8(3):513–538, 1982. doi: 10.2307/1972378.
- Economic Commission for Latin America and the Caribbean (ECLAC). Demographic observatory of latin america and the caribbean 2022: Population trends in latin america and the caribbean. Technical report, United Nations, Santiago, 2022.
- Jenny García and José Manuel Aburto. The impact of violence on Venezuelan life expectancy and lifespan inequality. *International Journal of Epidemiology*, 48(5):1593–1601, 2019. doi: 10.1093/ije/dyz061.
- Jenny García, Germán Correa, and Brenda Rousset. Trends in infant mortality in Venezuela between 1985 and 2016: A systematic analysis of demographic data. *The Lancet Global Health*, 7(4):e445–e457, 2019. doi: 10.1016/S2214-109X(18)30479-0.
- Paula E. Hollerbach, Sergio Díaz-Briquets, and Kenneth Hill. Fertility determinants in cuba. *International Family Planning Perspectives*, 10(1):12–20, 1984. doi: 10.2307/2948029.
- Ariel Karlinsky and Dmitry Kobak. Tracking excess mortality across countries during the COVID-19 pandemic with the World Mortality Dataset. *eLife*, 10:e69336, 2021. doi: 10.7554/eLife.69336.
- Nishant Kishore, Domingo Marqués, Ayesha Mahmud, Mathew V. Kiang, Irmay Rodriguez, Arlan Fuller, Peggy Ebner, Cecilia Sorensen, Fabio Racy, Jay Lemery, Leslie Maas, Jennifer Leaning, Rafael A. Irizarry, Satchit Balsari, and Caroline O. Buckee. Mortality in Puerto Rico after Hurricane Maria. *New England Journal of Medicine*, 379:162–170, 2018. doi: 10.1056/NEJMs1803972.
- Evelyn M. Kitagawa. Components of a difference between two rates. *Journal of the American Statistical Association*, 50(272):1168–1194, 1955. doi: 10.1080/01621459.1955.10501299.
- Ronald Lee and Andrew Mason. Is low fertility really a problem? Population aging, dependency, and consumption. *Science*, 346(6206):229–234, 2014. doi: 10.1126/science.1250542.
- Ron Lesthaeghe. The unfolding story of the second demographic transition. *Population and Development Review*, 36(2):211–251, 2010. doi: 10.1111/j.1728-4457.2010.00328.x.

- Mohamad Moslimani and Jeffrey S. Passel. Facts about the U.S. Cuban immigrant population. Technical report, Pew Research Center, 2024.
- William Msemburi, Ariel Karlinsky, Victoria Knutson, Serge Aleshin-Guendel, Somnath Chatterji, and Jon Wakefield. The WHO estimates of excess mortality associated with the COVID-19 pandemic. *Nature*, 613:130–137, 2023. doi: 10.1038/s41586-022-05522-2.
- ONEI. Census postponement and demographic programme notes, 2022. Official communications; census last completed 2012.
- ONEI. Effective population revision reported to the national assembly, 2024. End-2023 effective population 10,055,968.
- ONEI. Indicadores demográficos 2025 and anuario demográfico tables. <https://www.onei.gob.cu/poblacion-0>, 2025.
- Yasset Pérez-Riverol. Settled destinations ledger (aggregated public statistics). File `demographics/data/destinations_settled.csv` in Pérez-Riverol [2026d], 2026a.
- Yasset Pérez-Riverol. Model d sentinel-anchored monte carlo implementation. File `demographics/scripts/model_from_2021.py` in Pérez-Riverol [2026d], 2026b.
- Yasset Pérez-Riverol. Model e vital reconstruction (HSDS) — supplementary materials. File `demographics/manuscript/en/cuba-depopulation-2021-2026-supplement.pdf` in Pérez-Riverol [2026d], 2026c.
- Yasset Pérez-Riverol. Cubascience demographics companion repository. <https://github.com/ypriverol/cubascience>, 2026d.
- Samuel H. Preston, Patrick Heuveline, and Michel Guillot. *Demography: Measuring and Modeling Population Processes*. Blackwell, Oxford, 2001.
- United Nations, Department of Economic and Social Affairs, Population Division. World population prospects 2024. <https://population.un.org/wpp/>, 2024.
- U.S. Customs and Border Protection. Nationwide encounters: Cuba. <https://www.cbp.gov/newsroom/stats/nationwide-encounters>, 2024. Encounter events; not unique settled persons.
- World Food Programme. Cuba country brief / food-security assessments, 2024. Food insecurity and anaemia indicators cited as amplifying pathways, not as identified causes of death counts.